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Digital Design

→ Number system

Types :-

1. Binary
2. Octal
3. Decimal
4. Hexadecimal

{No. of digits = 0, 1, 2, ----, 9, A, B, C, D, E, F} (10 11 12 13 14 15) $\boxed{r=16}$

Binary: $\boxed{r=2}$ { No. of digits used 0 to $r-1$; 0 to $1 = 0, 1$ }

Octal: $\boxed{r=8}$ { No. of digits used = 0 to $r-1$ = 0 to 7 = 0, 1, 2, ----, 7 }

Decimal: $\boxed{r=10}$, No. of digits = 0 to 9 = 0, 1, 2, ----, 9

→ Determine the possible unknown base of a relation

Ex - $\sqrt{22} = 6$

Max^{m} value of digits = 6
base ≥ 7

8 → 0 to 7

$\sqrt{22} = 6$

$_{10_b} \quad _{0_b}$

Σ digits $\times b^{P.V}$

$\sqrt{2 \times b^0 + 2 \times b^1} = 6 \times b^0$

$2 + 2b = 36$

$b = \frac{34}{2} = 17$

4 bit = 1 nibble
8 bit = 1 byte = 2 nibble
16 bit = 1 word = 2 byte = 4 nibble

→ Number system conversions

1) Decimal to any other
2) Octal to binary
3) Binary to Octal
4) Any other to decimal
5) Hexa to binary
6) Binary to Hexa
7) Octal to Hexa
8) Hexa to Octal

1) Decimal to any other (DAO)

Q) 24_{10}

Binary:

$$\begin{array}{r|l} 2 & 24 \\ 2 & 12 - 0 \\ 2 & 6 - 0 \\ 2 & 3 - 0 \\ 2 & 1 - 1 \end{array}$$

$\boxed{24_{10} = 11000_2}$

Octal:

$$\begin{array}{r|l} 8 & 24 \\ & 3 - 0 \end{array}$$

$\boxed{24_{10} = 30_8}$

Hexa:

$$\begin{array}{r|l} 16 & 24 \\ & 1 - 8 \end{array}$$

$\boxed{24_{10} = 18_H = 18_{16}}$

Q) $24.25_{10} = X_2$

$0.25 \times 2 = 0.5 - 0$
$0.5 \times 2 = 1.0 - 1$
$0.00 \times 2 = 0.0 -$

& $24_{10} = 11000_2$

$\boxed{24.25 = 1100.01_2}$

Q) $23.9_{10} \rightarrow X_2$

$$\begin{array}{r|l} 2 & 23 \\ 2 & 11 - 1 \\ 2 & 5 - 1 \\ 2 & 2 - 1 \\ & 1 - 0 \end{array} = 10111_2$$

$0.9 \times 2 = 1.8 - 1$
$0.8 \times 2 = 1.6 - 1$
$0.6 \times 2 = 1.2 - 1$
$0.2 \times 2 = 0.4 - 0$
$= 1110$

$\boxed{23.9_{10} = 10111.1110_2}$

Q) $275.4_{10} \rightarrow X_8$ (Octal)

$$\begin{array}{r|l} 8 & 275 \\ 8 & 34 - 3 \\ & 4 - 2 \end{array} = 423_8$$

$0.4 \times 8 = 3.2 - 3$
$0.2 \times 8 = 1.6 - 1$
$0.6 \times 8 = 4.8 - 4$

$\boxed{275.4_{10} = 423.314_8}$

Q) $27.15_{10} \rightarrow X_8$

$$\begin{array}{r|l} 8 & 27 \\ & 3 - 3 \end{array} = 33_8$$

$0.15 \times 8 = 1.20 - 1$
$0.20 \times 8 = 1.6 - 1$
$0.6 \times 8 = 4.8 - 4$

$\boxed{27.15_{10} = 33.114_8}$

Q) $64.20_{10} \rightarrow X_{16}$

$$\begin{array}{r|l} 16 & 64 \\ & 4 - 0 \end{array} = 40_{16}$$

$0.20 \times 16 = 3.2 - 3$
$0.2 \times 16 = 3.2 - 3$

$\boxed{64.20_{10} = 40.333_{16}}$

NOTE* ---- $2^6, 2^5, 2^4, 2^3, 2^2, 2^1, 2^0$
----, 64, 32, 16, 8, 4, 2, 1

$(40)_{10} \rightarrow$ 1 0 1 0 0 0 $\rightarrow 101000_2$
$(34)_{10} \rightarrow$ 1 0 0 0 1 0 $\rightarrow 100010_2$
$(18)_{10} \rightarrow$ 1 0 0 1 0 $\rightarrow 10010_2$

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→ Any other number system into decimal (AOD)

Q) $1100_2 \to X_{10}$ (AOD)

3 2 1 0

$0\times2^0 + 0\times2^1 + 1\times2^2 + 1\times2^3$

$= 0+0+4+8$

$= 12$

$1100_2 \to 12_{10}$

OR

$1100_2 \to X_{10}$

$2^3\ 2^2\ 2^1\ 2^0$

8 4 2 1

$8+4+0+0 = 12$

$1100_2 \to 12_{10}$

ONLY FOR BINARY TO DECIMAL

Q) $1100.01_2 \to X_{10}$

3 2 1 0 -1 -2

$1\times2^{-2} + 0\times2^{-1} + 0\times2^0 + 0\times2^1 + 1\times2^2 + 1\times2^3 = 12.25_{10}$

Q) $11011011.1101_2 \to X_{10}$

7 6 5 4 3 2 1 0 -1 -2 -3 -4

$2^7\times1 + 2^6\times1 + 2^4\times1 + 2^3\times1 + 2^1\times1 + 2^0\times1 + 1\times2^{-1} + 1\times2^{-2} + 2^{-4}\times1$

$= 219.81_{10}$

→ Octal to decimal

Q) $12_8 \to X_{10}$

1 0

$2\times8^0 + 1\times8^1$

$= 10_{10}$

Conversion: Lower base to higher
↳ Value ↑ses
Higher base to lower
↳ Value ↓ses

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Q) $12.10_8 \to X_{10}$

1 0 -1 -2

$1\times8^1 + 2\times8^0 + 1\times8^{-1} + 0\times8^{-2}$

$= 10.12_{10}$

→ Hexadecimal to decimal

Q) $1F_H \to X_{10}$

1 0

$F\times16^0 + 1\times16^1$

$= 15\times1 + 16$

$= 31_{10}$

Q) $1F.10_H \to X_{10}$

1 0 -1 -2

$1\times16^1 + 15\times16^0 + 1\times16^{-1} + 0\times16^{-2}$

$= 31.06_{10}$

Q) $1FB.2A_H \to X_{10}$

2 1 0 -1 -2

$1\times16^2 + 15\times16^1 + 11\times16^0 + 2\times16^{-1} + 10\times16^{-2}$

$= 507.164_{10}$

→ Octal to binary & V.V

Octal	Binary 2^2 (4)	2^1 (2)	2^0 (1)
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Q) $64_8 \to X_2$

$6 \to 110$

$4 \to 100$

$64 \to 110100_2$

Q) $64.25_8 \to X_2$

$6 \to 110$

$4 \to 100$

$2 \to 010$

$5 \to 101$

$64.25_8 \to 110100.010101_2$

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Q) $1010.11_2 \rightarrow X_8$

421 421 . 421

$\underbrace{001}_{\text{Padded}}\ \underbrace{010}\ .\ \underbrace{110}_{\text{Padded}}$

$= 12.6_8$

→ **Hexadecimal to Binary & V.V**

Hexa ($16=2^4$)	Binary (2^3 2^2 2^1 2^0 / 8 4 2 1)
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1
A) 10	1 0 1 0
B) 11	1 0 1 1
C) 12	1 1 0 0
D) 13	1 1 0 1
E) 14	1 1 1 0
F) 15	1 1 1 1

Q) $64.25_H \rightarrow X_2$

$= 01100100.00100101_2$

Q) $10010.1101_2 \rightarrow X_H$

$\underbrace{0001}_{\text{Pad}}\underbrace{0010}.\underbrace{1101}$

$= 12.13_2$

$= 12.D_2$

Q) $ABF.2F_H \rightarrow X_2$

101010111111.00101111_2

Good Write

8 4 2 1

4 2 1

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→ **Hexa to Octal & V.V**

(First we need to convert this into binary & then binary to octal)

Q) $AF.2B_H \rightarrow X_8$

Binary → $\underbrace{010}_{\text{Pad}}101111.001010110_2$ (Pad)

Octal → 257.126_8

Q) $72.1_8 \rightarrow X_H$

Binary → 111010.001_2

Hexa → $\underbrace{0011}_{\text{Pad}}1010.\underbrace{0010}_{\text{Pad}}{}_2$

$= 3A.2_H$

Q) $1217_8 \rightarrow X_{16}$

Binary → 001010001111

Hexa → $28F_{16}$

Q) $12C9_{16} \rightarrow X_8$

$\underbrace{00000}_{\text{Pad}}1001011001001_2$

011311_8

→ How many bits are required to represent 16 digit decimal number

$2^n > \text{number}$

$n = \text{No. of digits}$

$2^n > 16$ digit decimal no.

$2^n > \underbrace{99\ldots\ldots999}_{16}$

$2^n > 10^{16}$

$16 < \log_{10} 2^n$

$n > \frac{16}{0.3} \Rightarrow \boxed{n > 53.156}$

$99 \approx 10^2$
$999 \approx 10^3$
$9999 \approx 10^4$

Good Write

Q) How many bits are required to represent $(4085)_{10}$

4085_{10}

$2^n > \text{number}_{10}$

n = No. of binary bits

$2^n > 4085$ $\qquad 2^{11} < 4085$

$2^{12} > 4085$ $\qquad 2^{12} > 4085$

$\boxed{n = 12}$

Analog signal → ADC → Processor → DAC → Analog

Binary no. ⇒ $\underbrace{100}_{3 \text{ bits}}$ $\qquad \underbrace{1010}_{4 \text{ bits}}$ = nibble

$\underbrace{1001}\,\underbrace{0110}$

8 bits = 2 nibble = 1 byte

32 kbps → kilo bit per sec

4 KBps → kilo Byte per sec

n-bits can have 2^n possible combination

→ Positional number system

$(786)_{10} = 7 \times 100 + 8 \times 10 + 6 \times 1$

Most significant digit = 7 (MSD)

Least " " = 6 (LSD)

→ Complement

r's complement $\qquad$ (r-1)'s complement

$N = (785.32)_{10}$

No. of integer digits = 3 = n

$\boxed{\text{r's complement} = r^n - N}$

10's comp. of $N = 10^3 - 785.32$

$= (214.68)_{10}$

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Q) 2's complement of $(101100)_2$

$\Rightarrow 2^6 - (101100)_2$
$\Rightarrow 2^6 - 44$
$\Rightarrow (20)_{10}$
$\Rightarrow (010100)_2$

101100
5 4 3 2 1 0
32 + 8 + 4 = 44

Q) 2's complement of $(0.0110)_2$
$n = 0$

$2^0 - (0.0110)_2$ (place values −1 −2 −3 −4)
$2^0 - (2^{-2} + 2^{-3})$
$1 - (0.375)$
$= (0.625)_{10}$

0.625 × 2 = 1.25	1
0.25 × 2 = 0.50	0
0.50 × 2 = 1.00	1
0.00 × 2 = 0.00	0

$\Rightarrow (0.1010)_2$

Q) → (r−1)'s complement
$N = (785.32)_{10}$
No. of integer digits = n = 3
No. " decimal = m = 2

$\boxed{(r-1)\text{'s complement} = r^n - r^{-m} - N}$

9's ~~10's~~ complement $= 10^3 - 10^{-2} - 785.32$
$= 1000 - 0.01 - 785.32$
$= (214.67)_{10}$

OR

999.99
−785.32
214.67

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Q) $(52520)_{10}$

9's complement
99999
− 52520
$(47479)_{10}$

10's complement
$(47479) + 10^{-0}$
$= (47480)_{10}$

$\boxed{r\text{'s complement} = (r-1)\text{'s complement} + r^{-m}}$

1's complement
$(0.0110)_2$

0.1111
−0.0110
0.1001_2

2's complement

$(0.1001)_2 + 2^{-4}$
$(0.1001)_2 + (0.0001)_2$ (place values −1 −2 −3 −4)
$= (0.1010)_2$

→ Subtraction with r's complement

(M − N)
Minuend — Subtrahend
n

72532 − 03250

Base = 10
r's complement $= 10^5 - 3250$
$= 96750$

 (1)
 72532
+96750
 (1) 69282
Carry

Carry occurs → discard

$\boxed{\text{Result} \rightarrow (69282)_{10}}$

Q) 3250 - 72532

r's complement = $10^5 - 72532$
= 27468

$$\begin{array}{r} 03250 \\ +27468 \\ \hline 30718 \end{array}$$

CY = 0

r's complement = $10^5 - 30718$
= $-(69282)_{10}$

→ Subtraction using (r-1)'s complement

Q) 72532 - 03250

Base = 10

9's complement = 99999 - 03250 = 96749

$$\begin{array}{r} 72532 \\ +96749 \\ \hline 69281 \end{array}$$

$\boxed{CY = 1}$

$$\begin{array}{r} 69281 \\ +1 \\ \hline (69282)_{10} \end{array}$$ Ans

Q) 3250 - 72532

Base = 10

9's complement = 99999 - 72532 = 27467

$$\begin{array}{r} 27467 \\ +03250 \\ \hline 30717 \end{array}$$

$\boxed{CY = 0}$

9's complement

$$\begin{array}{r} 99999 \\ -30717 \\ \hline -(69282)_{10} \end{array}$$ Ans

Q) $(0110)_2 - (1000)_2$

↑ Minuend ↑ subtrahend

1's complement = 1111 - 1000 = 0111

2's complement = 1's complement + r^{-m}
= $(0111) + 2^{-0}$
= $(0111) + 1$

$$\begin{array}{r} 0111 \\ 0001 \\ \hline 1000 \end{array}$$

Using 1's complement

$$\begin{array}{r} 0110 \\ +0111 \\ \hline (1101)_2 \end{array}$$

$\boxed{CY = 0}$

1's complement = $-(0010)_2 = (-2)_{10}$

Using 2's complement

$$\begin{array}{r} 0110 \\ +1000 \\ \hline 1110 \end{array}$$

$\boxed{CY = 0}$

1's comp. = 0001

2's comp. = $0001 - 2^{-0}$

$$\begin{array}{r} = 0001 \\ -0001 \\ \hline -(0010)_2 \end{array}$$ $= (-2)_{10}$

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Representation of binary no.

1) Sign magnitude form

n bit binary no.

MSB → Sign bit → 1 -Ve no.
→ 0 +Ve no.

(n-1) bits → magnitude

Q) $(1\,0\,1\,1\,0\,1)_2$ (weights: 16 8 4 2 1)

Sign ⇒ $(-13)_{10}$

3 bits → 8 comb^n → 0 to 7
n bits → 2^n → 0 to $(2^n - 1)$
(n-1) bits → $(2^{n-1}-1)$ to $(2^{n-1}-1)$

2) 2's complement representation

- +Ve no. → As it is
- -Ve no. → Take 2's complement & then express.

Q) $(45)_{10}$

⇒ 1 0 1 1 0 1
32 16 8 4 2 1

$(101101)_2$

Q) $(-45)_{10}$

2's complement of $(101101)_2$

1's complement = $(010010)_2$

2's " = $(010010) + 2^{-0}$

010010
000001
$(010011)_2$

$\boxed{\text{Range :- } (-2^{n-1}) \text{ to } (2^{n-1}-1)}$

4 bits ⇒ n = 4
-8 to 7

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3) 1's complement representation

- +Ve no. → As it is
- -Ve no. → Take 1's complement & then express

4) Overflow

Q) 9 + 4

8 4 2 1
Sign = 0 1 0 0 1
0 0 1 0 0
0 1 1 0 1

+Ve no. $(+13)_{10}$

Q) 9 − 4 ⇒ 9 + (−4)

01001
~~1's 00100~~
~~1's 10110~~ 4 ⇒ 00100
1's ⇒ 11011
2's ⇒ 11011
+1
11100

01001
11100
Cy = 1 00101

+Ve $(+5)_{10}$

Q) (−9) + 4

01001
1's ⇒ 10110
2's ⇒ 10110
+1
10111

10111
00100
11011

Sign = 1

1011
1's : 0100
2's : $(0101)_2$
$(-5)_2$

Overflow

Q) 9 + 7

```
  0 1 0 0 1
  0 0 1 1 1
 -----------
 (1) 0 0 0 0
  ↑
 Sign bit
```

When the result of two nos. is beyond the range of representation then the sign bit is incorrect

→ **Binary Codes**

1) **Binary Coded Decimal (BCD)**

No. of bits required = m
Such that $2^m \geq M$
M is no. of symbols

Decimal	b_4 (8)	b_3 (4)	b_2 (2)	b_1 (1)	Unused
					1010
0	0	0	0	0	1011
1	0	0	0	1	1100
2	0	0	1	0	1101
3	0	0	1	1	1110
4	0	1	0	0	1111
5	0	1	0	1	
6	0	1	1	0	Invalid
7	0	1	1	1	
8	1	0	0	0	
9	1	0	0	1	

Q) $(756.24)_{10}$
$(0111\ 0101\ 0110.\ 0010\ 0100)_2$

→ **BCD addition**

```
   25        0010   0101
 + 13      + 0001   0011
 (38)10      0011   1000
```

Sum ≤ 9, final carry = 0 → Answer is correct

```
   25        0010   0101
 + 16        0001   0110
 (41)10      0011   1001
```

Sum ≤ 9, final carry = 1 → Answer is wrong add 6 (0110)

Sum > 0, final carry = 0 → Answer is wrong (add 0110)

8421
= 11 > 9
(Invalid)
Not allowed in BCD

Add 6

```
   0011   1011
 +        (0110) = 6
   0100   0001
```

Q)

```
   679.6        0110   0111   1001 . 0110
 + 536.8      + 0101   0011   0110 . 1000
 (1216.4)10     1011   1010   1111 . 1110
                11>9   10>9   15>9   14>9

                1011   1010   1111 . 1110
              + 0110   0110   0110 . 0110  ⇒ 6
 0001 ← CY=1    0010   0001   0110 . 0100
                (1216.4)10
```

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→ BCD subtraction

$$\begin{array}{r} 206.8 \\ -147.6 \\ \hline (59.2)_{10} \end{array}$$

(r−1)'s complement
Using 9's complement

$$\begin{array}{r} 999.9 \\ -147.6 \\ \hline 852.3 \end{array}$$

(206.8)	0010	0000	0110	. 1000
(852.3) +	1000	0101	0010	. 0011
	1010	0101	1000	. 1011
	+ 0110			0110
~~0001~~ ← $\boxed{CY = 1}$	0000	0101	1001	. 0001
				0001
				0010

~~$(1059.1)_{10}$~~

$(059.2)_{10}$
$(59.2)_{10}$

2) Excess-3 code (XS-3)

Decimal	XS-3	Binary b_3 (8)	b_2 (4)	b_1 (2)	b_0 (1)
0	3	0	0	1	1
1	4	0	1	0	0
2	5	0	1	0	1
3	6	0	1	1	0
4	7	0	1	1	1
5	8	1	0	0	0
6	9	1	0	0	1
7	10	1	0	1	0
8	11	1	0	1	1
9	12	1	1	0	0

Unused

0000	(0)
0001	(1)
0010	(2)
1101	(13)
1110	(14)
1111	(15)

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3) Gray Code (Reflective Code)

Binary (1) 0 0 0 1
(2) 0 0 1 0

Gray code is a type of binary code in which there is a change of only 1-bit in successive numbers.

2-bits 0, 1, 3, 2

00, 01, 11, 10

3-bits − 0, 1, 3, 2, 6, 7, 5, 4

000 → 001 → 011 → 010 → 110 → 111
100 ← 101

4-bits 0, 1, 3, 2, 6, 7, 5, 4, 12, 13, 15, 14, 10, 11, 9, 8

0000 → 0001 → 0011 → 0010 → 0110 → 0111 → 0101 → 0100
1000 ← 1001 ← 1011 ← 1010 ← 1110 ← 1111 ← 1101 ← 1100

→ Binary to Gray

$G_m = B_m$ − (MSB) → XOR
$G_{m-1} = B_m \oplus B_{m-1}$
$G_{m-2} = B_{m-1} \oplus B_{m-2}$

1 0 0
↓ ↓ ↓
1 1 0

XOR diff i/p → 1
same i/p → 0

→ Gray to Binary

$B_m = G_m$ (MSB)
$B_{m-1} = B_m \oplus G_{m-1}$
$B_{m-2} = B_{m-1} \oplus G_{m-2}$

1 1 1
↓ ↓ ↓
1 0 1

) Parity Code

	b_3	b_2	b_1	b_0	P	
OPG	0	1	1	0	1	⇒ 3
	1	0	0	0	0	⇒ 1
EPG	0	1	1	0	0	⇒ 2
	1	1	1	0	1	⇒ 4

Parity checker

EPC · o/p → error
data no. of 1's → odd

OPC o/p → error
data no. of 1's → even

→ Logic Gates

1) Buffer
2) Inverter (NOT)
3) AND
4) OR
5) NAND
6) NOR
7) XOR
8) XNOR

→ Buffer

1) Bistate buffer
Two states → 0 & 1

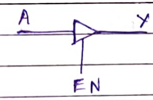
ⓐ

A	Y
0	0
1	1

ⓑ Logic symbol

2) Tristate buffer

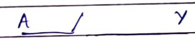
3 states 0, 1, Hi-Z
Hi-Z → High impedance (Open circuit)



If EN = 1 → Bistate buffer

A	Y
0	0
1	1

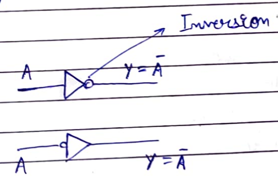
If EN = 0 ; Y = unknown



Y is not connected to A.

→ Inverter (NOT gate)

A	Y
0	1
1	0



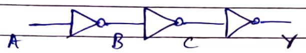
→ Cascading of NOT gate

① Even no. of NOT gates

A	B	Y
0	1	0
1	0	1

$Y = \bar{A}$
$B = \bar{A}$
$Y = \bar{B} = \bar{\bar{A}} = A$ (Buffer)

② Odd no. of NOT gates

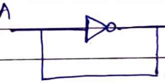


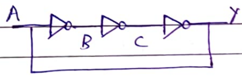
$Y = \bar{A}$ (inverter)

A	B	C	Y
0	1	0	1
1	0	1	0

→ NOT gate with feedback

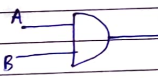
① Odd no. of NOT gate





Square-wave generator
Clock generator
Astable multi-vibrator
Ring oscillator

→ AND gate

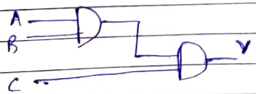


Two input AND gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

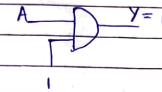
AND operation is commutative
$\boxed{AB = BA}$

AND operation is associative
$(AB)C = A(BC)$



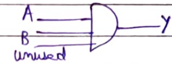
→ Enable & disable

Disabled

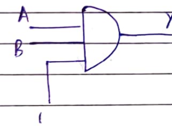


Enabled

Unused input

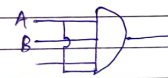


M-1 : Connect unused input to V_{cc}



$Y = ABC$
$= AB(1)$
$= AB$ (Best method)

M-2 - Connect unused i/p to existing i/p



$Y = (AB)A$
$= (AB)$

Drawback → Fanout ↓

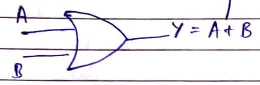
M-3 - Unused i/p is left floating
↳ Only possible in TTL logic Family
↳ floating i/p → '1'

DATE: __/__/____
PAGE: ______

OR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

+ sign does not show add^n



2 input OR gate

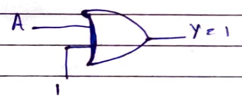
Commutative law

$A + B = B + A$

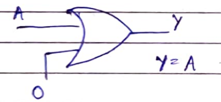
Associative law

$(A+B)+C = A+(B+C)$

→ **Enable & disable input**

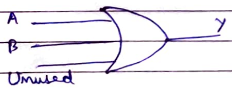


Disabled gate

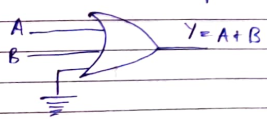


Enabled gate

→ **Unused input**



M-1 : Connect unused input to ground



M-2 : Connect unused input to used input

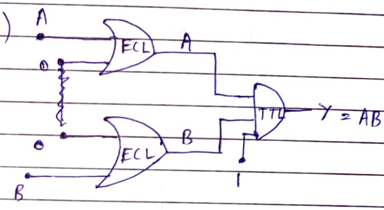
$Y = (A+B)+A$
$= A+A+B$
$= A+B$

Good Write

DATE: __/__/____
PAGE: ______

M-3 - unused i/p is left floating
↳ ECL logic family
↳ floating i/p = '0'
(Noise margin) ↓

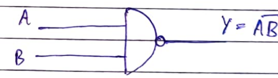
Q)



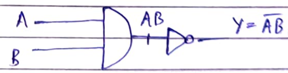
→ **NAND gate** (If both inputs are true then output will be false)

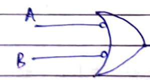
NOT + AND = NAND

A	B	AB	Y
0	1	0	1
1	0	0	1
1	1	1	0
0	0	0	1



2 input NAND gate





$Y = \bar{A} + \bar{B}$
$Y = \overline{AB}$ (De-Morgan's law)

Bubbled OR gate
= NAND gate

Commutative law

$\overline{AB} = \overline{BA}$

Associative law

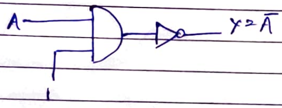
$\overline{(\overline{AB})C} \neq \overline{A(\overline{BC})}$

Does not follow

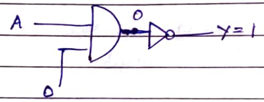
3 input NAND ≠ 2 input NAND

Good Write

→ Enable & disable



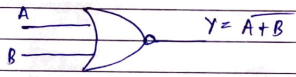
Enabled gate

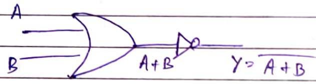


disabled gate

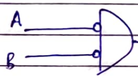
→ NOR gate

NOT + OR = NOR





A	B	A+B	Y	$\bar{A}$	$\bar{B}$	
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0



$Y = \bar{A}.\bar{B}$
$= (\overline{A+B})$

bubbled AND gate

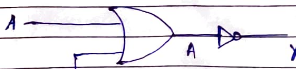
Commutative law

$(\overline{A+B}) = (\overline{B+A})$

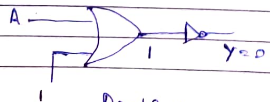
Associative law

$(\overline{A+B})C \neq \overline{A+(B+C)}$

→ Enabled & disabled



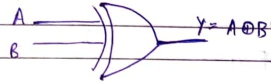
Enabled



Disabled

→ Ex-OR gate (XOR)

Exclusive disjunction



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Y = 1 if $A \neq B$
↳ inequality detector

$\boxed{Y = \bar{A}B + A\bar{B}}$ OR $Y = (A+B)(\bar{A}+\bar{B})$: $\boxed{POS}$
$\boxed{SOP}$
$= A\bar{A} + \bar{A}B + A\bar{B} + B\bar{B}$
$= \bar{A}B + A\bar{B}$

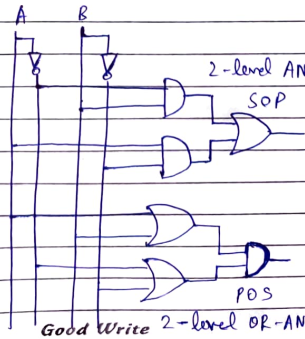
XOR gate gives an output 1 if odd no. of 1's are present at i/p

Commutative law

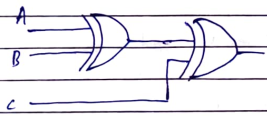
$A \oplus B = B \oplus A$

Associative

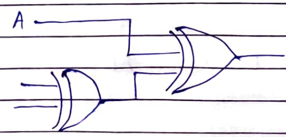
$(A \oplus B) \oplus C = A \oplus (B \oplus C)$



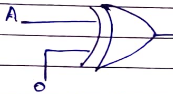
2-level OR-AND



3 input XOR = 2 i/p XOR

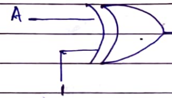


→ Enable & disable input



$Y = A\bar{B} + \bar{A}B$
$= A(\bar{0}) + (\bar{A})(0)$
$= A$ (buffer)

Disable



$Y = A\bar{B} + \bar{A}B$
$= A(\bar{1}) + \bar{A}1$
$= A(0) + \bar{A}1$
$= \bar{A}$ (inverter)

Enable

→ Properties

1) $A \oplus 0 = A$
2) $A \oplus 1 = \bar{A}$
3) $A \oplus A = 0$
4) $A \oplus \bar{A} = 1$

5) $A \oplus A \oplus A$ (Odd no.)
$(A \oplus A) \oplus A$
$0 \oplus A$
A

6) $A \oplus A \oplus A \oplus A$ (Even)
$(A \oplus A) \oplus (A \oplus A)$
$0 \oplus 0$
0

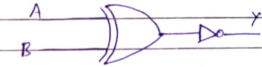
if $A \oplus B = C$
then, $A \oplus C = B$
$B \oplus C = A$

3-input XOR gate

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	~~A~~ 1

→ Ex-NOR (XNOR)

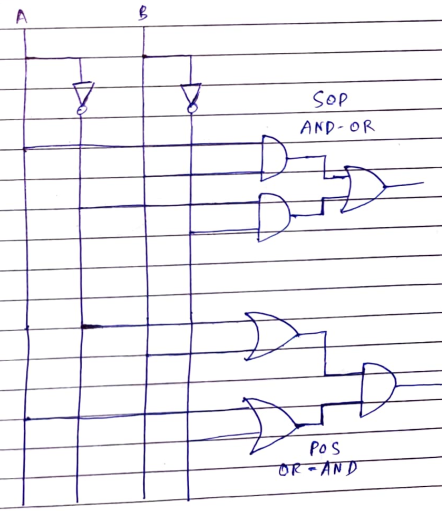
Exclusive NOR



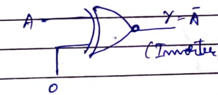
A	B	$A \oplus B$	Y
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

$Y = \overline{A \oplus B} = A \odot B$
$= AB + \bar{A}\bar{B}$ (SOP)
$= (\bar{A} + B)(A + \bar{B})$ (POS)

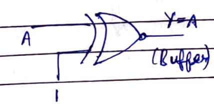
Y is 1 when $A = B$
↳ Equality detector



Disable



Enable



Properties

1) $A \odot 0 = \bar{A}$
2) $A \odot 1 = A$
3) $A \odot A = 1$
4) $A \odot \bar{A} = 0$

5) $A \odot A \odot A$ (Odd)
1

6) $A \odot A \odot A \odot A$ (Even)
A

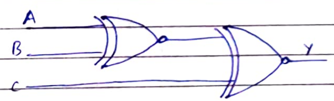
Commutative law
$A \odot B = B \odot A$

Associative law
$(A \odot B) \odot C = A \odot (B \odot C)$

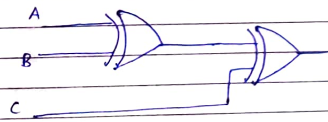
$A \odot B = \overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B}$

$A \oplus B = \overline{A \odot B} = \bar{A} \odot B = A \odot \bar{B}$

3 input XNOR $(A \odot B \odot C)$



For odd no. of input XNOR ~~not~~ gate will behave like a XOR gate



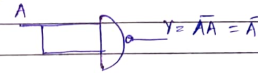
A	B	C	XOR	XNOR
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

→ **Universal gate**

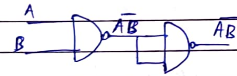
NAND NOR

NAND as universal gate

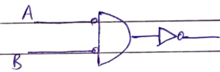
① **Inverter**

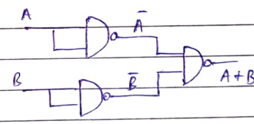


② **AND**

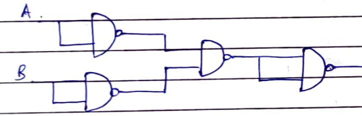


③ **OR** $(Y = A + B)$

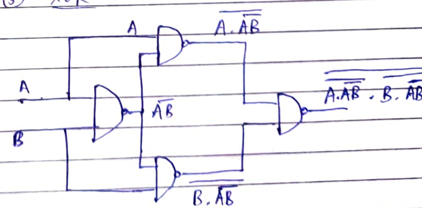




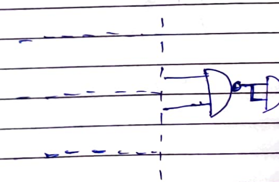
④ **NOR**



⑤ **XOR**



⑥ **XNOR**



Replace  with  for NOR as universal gate

& A.B by A+B

→ ~~Bot~~ Boolean Algebra

Boolean variables
↳ 2 possible values

HIGH	LOW
TRUE	FALSE
ON	OFF
logic 1	logic 0

Axioms of Boolean Algebra

① NOT operation

$Y = \bar{A}$

$\bar{1} = 0$

$\bar{0} = 1$

$\bar{\bar{A}} = A$

A	Y
0	1
1	0

② AND operation (.)

$1.0 = 0$ $\quad 0.1 = 0$

$0.0 = 0$ $\quad 1.1 = 1$

$A.A = A$

$A.1 = A$

$A.0 = 0$

A	A.A = A
0	0
1	1

$A.\bar{A}.A.A$ (n-times) $= 0$

A	$\bar{A}$	$A.\bar{A}$
0	1	0
1	0	0

③ OR operation (+)

$0+0 = 0$ $\quad 1+0 = 1$

$0+1 = 1$ $\quad 1+1 = 1$

$A+A = A$ $\quad A+\bar{A} = 1$

$A+1 = 1$ $\quad A+0 = A$ $\quad \boxed{1+f(A,B,C) = 1}$

A	A+A
0	0
1	1

Commutative law

AND $\quad AB = BA$

OR $\quad A+B = B+A$

All other logic gates are commutative

Implication gate → $A \to B = \bar{A} + B \neq \bar{B} + A$ (Not commutative)

Inhibition gate → $A/B = A\bar{B} \neq \bar{A}B$ (Not commutative)

Associative law

OR $\quad A+(B+C) = (A+B)+C$

AND $\quad A(BC) = (AB)C$

NAND, NOR are not associative

Distribution theorem

$\boxed{(A+B)(A+C) = A+BC}$

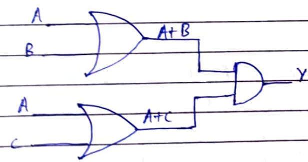
$A.A + AC + BA + BC$

$A + AC + AB + BC$

$BC\ \cancel{A} + A(C+B+1)$

$A.1 + BC$

$A + BC$



=

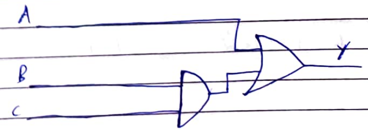
$(A+B+C)(A+B+D)$

$X = A+B$

$(X+C)(X+D)$

$= X+CD$

$= A+B+CD$



$f(A,B,C) = (A+B)(A+\bar{B})(\bar{A}+C)(\bar{A}+\bar{C})$

$= (A+B\bar{B})(\bar{A}+C\bar{C})$

$= (A+0)(\bar{A}+0)$

$= 0 + A\bar{A}$

$= \cancel{1+1}\ 0$

$= \cancel{1}$

$f(A,B) = \cancel{A+B}^{AB} + A\bar{B}$

$= A(B+\bar{B})$

$= A.1 = A$

NAND gates = 0

$f(A,B,C,D) = A + AB + ABC + ABCD$

$= A(1+B+BC+BCD)$

$= A.1 = A$

NAND gates = 0

→ Consensus theorem

$\boxed{AB + \bar{A}C + \underbrace{BC}_{\text{Redundant}} = AB + \cancel{B}\bar{A}C}$ (SOP)

$AB + \bar{A}C + BC(1)$

$= AB + \bar{A}C + BC(A+\bar{A})$

$= AB + \bar{A}C + ABC + \bar{A}BC$

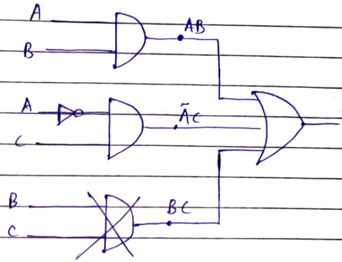
$= AB(1+C) + \bar{A}C(1+B)$

$= AB + \bar{A}C$

$\boxed{(A+B)(\bar{B}+C)(\bar{A}+C) = (A+B)(\bar{B}+C)}$ (POS)

$A\bar{B} + AB + \bar{A}C = A(\bar{B}+B) + \bar{A}C = A+\bar{A}C = A+C$

Consensus ✗



Redundant gate

$f = XY + \bar{X}W + WUYZP$

↘ Redundant

$= XY + \bar{X}W$

→ Redundant literal rule (RLR)

$\boxed{X + \bar{X}Y = X + Y}$ $\boxed{\bar{X} + XY = \bar{X}Y}$

If $X=1$, $1+0Y=1$, $1+Y=1$

If $X=0$, $0+1Y=Y$, $0+Y=Y$

→ De·Morgan's theorem

$\overline{ABC} = \bar{A}+\bar{B}+\bar{C}$

$\overline{A+B+C} = \bar{A}.\bar{B}.\bar{C}$

$\overline{(A+B).C} = \overline{(A+B)} + \bar{C}$

$= \bar{A}.\bar{B} + \bar{C}$

→ Shannon's expansion theorem

$$f(A,B,C,D) = A.f(1,B,C,D) + \bar{A} f(0,B,C,D)$$
$$f(A,B,C,D) = [A + f(0,B,C,D)][\bar{A} + f(1,B,C,D)]$$

→ Transposition theorem

$$\boxed{AB + \bar{A}C = (\bar{A}+B)(A+C)}$$

→ Dual of a logic expression

OR ⟷ AND

1 ⟷ 0

$$AB + \bar{A}C + BC = AB + \bar{A}C$$
$$(A+B).(\bar{A}+C).(B+C)$$
$$= (A+B).(\bar{A}+C)$$

→ Positive & Negative logic

Positive

AND

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Negative

AND

A	B	Y
1	1	1
1	0	1
[illegible]	[illegible]	1
0	0	0

OR

A	B	Y
1	1	1
1	0	0
0	1	0
0	0	0

Complement

1) Complement each variable
2) AND ⟷ OR
3) 1 ⟷ 0

Q) $f = AB + BC + AC$

$f^D = (A+B).(B+C).(A+C)$

$\bar{f} = \overline{AB+BC+AC}$

$= \overline{AB}.\overline{BC}.\overline{AC}$

$= (\bar{A}+\bar{B})(\bar{B}+\bar{C})(\bar{A}+\bar{C})$

Self-Dual

$f^D = (A+B)(AB+AC+BC+C)$

$= (A+B)(AB+C(1+A+B))$

$= (A+B)(AB+C)$

$= AB + AB + BC + AC$

$= AB + BC + AC$

Q) $f = AB\bar{C} + \bar{A}BC + A\bar{B}C$

$f^D = (A+B+\bar{C}).(\bar{A}+B+C).(A+\bar{B}+C)$

$f^C = (\bar{A}+\bar{B}+C)(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{C})$

→ Minterm & Maxterm

Decimal	A (2^1)	B (2^0)		Y (XOR)
0	0	0	$m_0 = \bar{A}\bar{B}$	0
1	0	1	$m_1 = \bar{A}B$	1
2	1	0	$m_2 = A\bar{B}$	1
3	1	1	$m_3 = AB$	0

Minterm is 1 for a particular combination of input variable

~~Minterm~~ $Y = m_1 + m_2$

$= \Sigma m(1,2)$

Decimal	A (2^2)	B (2^1)	C (2^0)		Y (XOR)
0	0	0	0	$m_0 = \bar{A}\bar{B}\bar{C}$	0
1	0	0	1	$m_1 = \bar{A}\bar{B}C$	1
2	0	1	0	$m_2 = \bar{A}B\bar{C}$	1
3	0	1	1	$m_3 = \bar{A}BC$	0
4	1	0	0	$m_4 = A\bar{B}\bar{C}$	1
5	1	0	1	$m_5 = A\bar{B}C$	0
6	1	1	0	$m_6 = AB\bar{C}$	0
7	1	1	1	$m_7 = ABC$	1
8				$m_8 =$	
				$m_9 =$	

For n-input variables $= 2^n$ minterms

Each logic expression can be expressed as sum of the minterms

$Y = m_1 + m_2 + m_4 + m_7$

$= \Sigma m(1,2,4,7)$ [SOP]

Maxterms

Each maxterm is a sum term that contains all variables either in complemented or uncomplemented form.
Each maxterm is 0 for 1 particular combination of input variable.

Decimal	A (2^1)	B (2^0)		Y
0	0	0	$M_0 = A+B$	0
1	0	1	$M_1 = A+\bar{B}$	1
2	1	0	$M_2 = \bar{A}+B$	1
3	1	1	$M_3 = \bar{A}+\bar{B}$	0
~~4~~				

n-input variables $= 2^n$ maxterms

$Y = M_0 . M_3$
$= \prod M(0,3)$

Decimal	A	B	C		Y
0	0	0	0	$M_0 = A+B+C$	0
1	0	0	1	$M_1 = A+B+\bar{C}$	1
2	0	1	0	$M_2 = A+\bar{B}+C$	1
3	0	1	1	$M_3 = A+\bar{B}+\bar{C}$	0
4	1	0	1	$M_4 = \bar{A}+B+\bar{C}$	1
5	1	~~1~~ 0	~~0~~ 1	$M_5 = \bar{A}+B+\bar{C}$	0
6	1	1	0	$M_6 = \bar{A}+\bar{B}+C$	0
7	1	1	1	$M_7 = \bar{A}+\bar{B}+\bar{C}$	1

$Y = M_0 M_3 M_5 M_6$
$= \prod M(0,3,5,6)$
[POS]

With 'n' input variable no. of expressions possible ?

$$\boxed{2^{(2^n)}}$$

Sum of all the minterms is 1

$$\boxed{\sum_{i=0}^{2^n-1} m_i = 1}$$

Product of all the maxterms is 0

$$\boxed{\prod_{i=0}^{2^n-1} M_i = 0}$$

$$\boxed{m_i^D = M_{2^n-1-i}} \qquad \boxed{\overline{m_i} = M_i}$$

→ Forms of Boolean Function

1) Sum of product (SOP)
Disjunction Normal form
$f(A,B,C) = AB + \bar{A}C$

2) Product of Sum (POS)
Conjunctive Normal form
$f(A,B,C) = (A+B)(\bar{A}+C)$

3) Standard SOP
Disjunctive Canonical form

$f(A,B,C) = AB.1 + \bar{A}C.1$
$= (C+\bar{C})AB + \bar{A}C(B+\bar{B})$
$= \underset{001}{\bar{A}\bar{B}C} + \underset{011}{\bar{A}BC} + \underset{110}{AB\bar{C}} + \underset{111}{ABC}$
$= \sum m(1,3,6,7)$

4) Standard POS
Conjunctive Canonical form

$f(A,B,C) = (A+B+0)(\bar{A}+C+0)$
$= (A+B+C\bar{C})(\bar{A}+C+B\bar{B})$
$= \underset{000}{(A+B+C)}\underset{001}{(A+B+\bar{C})}\underset{110}{(\bar{A}+\bar{B}+C)}\underset{100}{(\bar{A}+B+C)}$

$f = \prod M(0,1,4,6)$

SOP ⟷ POS

$f = \Sigma m(1,3,6,7)$

$\bar{f} = \Sigma m(0,2,4,5)$

$\bar{f} = m_0 + m_2 + m_4 + m_5$

$f = \overline{m_0 + m_2 + m_4 + m_5}$

$= \bar{m}_0 . \bar{m}_2 . \bar{m}_4 . \bar{m}_5$

$= M_0 M_2 M_4 M_5$

$= \Pi M(0,2,4,5)$

Q) $f(A,B,C,D) = AB$

$= AB(C+\bar{C})(D+\bar{D})$

$= AB(\bar{C}\bar{D} + \bar{C}D + C\bar{D} + CD)$

$= \underset{1100}{AB\bar{C}\bar{D}} + \underset{1101}{AB\bar{C}D} + \underset{1110}{ABC\bar{D}} + \underset{1111}{ABCD}$

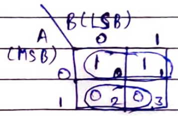
$= \Sigma m(12,13,14,15)$

→ Karnaugh Map (K-map)

2-variable K-map

Decimal	A	B	Y	m	M
0	0	0	1	$m_0 = \bar{A}\bar{B}$	$M_0 = A+B = \bar{m}_0$
1	0	1	1	$m_1 = \bar{A}B$	$M_1 = A+\bar{B} = \bar{m}_1$
2	1	0	0	$m_2 = A\bar{B}$	$M_2 = \bar{A}+B = \bar{m}_2$
3	1	1	0	$m_3 = AB$	$M_3 = \bar{A}+\bar{B} = \bar{m}_3$

n-variable K-map $= 2^n$ cells



$Y = m_0 + m_1$

$= \bar{A}\bar{B} + \bar{A}B$

$= \bar{A}(B+\bar{B})$

$= \bar{A}$

SOP

When terms are combined common literal is taken

$Y = M_2 . M_3$

$= (\bar{A}+B)(\bar{A}+\bar{B})$

$= \bar{A} + \bar{A}\bar{B} + \bar{A}B + B\bar{B}$

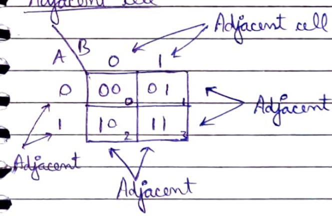
$= \bar{A} + \bar{A}(\bar{B}+B) + 0$

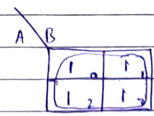
$= \bar{A} + \bar{A}.1 = \bar{A}$

→ Grouping of adjacent cells

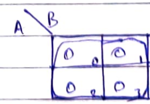
Two cells are said to be adjacent if they lie at unit dist., only one bit changes from one cell to the other.

Adjacent cell





$f = m_1 + m_2 + m_3 + m_0 = 1$



$f = M_0 M_1 M_2 M_3 = 0$

When 2^n ~~n~~ adjacent cells are grouped then n-variables are eliminated.

Group of 2 adjacent cells → Pair
" " 4 " " → Quad
" " 8 " " → Octate

→ **Don't care condition**

The output of a logic expression can have three possible values 0, 1 & X (Don't care).

Q) $f(A,B) = \Sigma m(0,1) + d(2)$

A \ B	0	1
0	1_0	1_1
1	X_2	$_3$

No need to include

All 1's must be covered while they are in SOP
& 0's " " " " " " POS

$f = \bar{A} + \bar{B}$ (Not minimized)
$f = \bar{A}$

Q) $f(A,B) = \Sigma m(0,1) + d(2,3)$

A \ B	0	1
0	1_0	1_1
1	X_2	X_3

$f = 1$

Q) $f(A,B) = \Sigma m(0,3) + d(2)$

A \ B	0	1
0	1_0	$_1$
1	X_2	1_3

$f = m_0 + m_3 = \bar{A}\bar{B} + AB = A \odot B$ 4 literals ✗

$\boxed{f = \bar{B} + A}$

2 literals ✓

3- Variable K-map

$f(A, B, C)$
↑ MSB ↑ LSB

3 variable = 8 minterms
= 8 maxterms

8 cells in K-map

"Gray code"

A \ B C	00	01	11	10
0	0	1	3	2
1	4	5	7	6

AB \ C	0	1
00	0	1
01	2	3
11	6	7
10	4	5

K-map roll off

00 → 01
↓
10 ← 11

Q) $f(A,B,C) = \Sigma m(0,1,2,3)$

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	1_0	1_1	1_2	1_3
A 1	4	5	7	6

$f = \bar{A}$

Q) $f(A,B,C) = \Sigma m(1,3,6,7)$

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	$_0$	1_1	1_3	$_2$
A 1	$_4$	$_5$	1_7	1_6

$f(A,B,C) = \bar{A}C + AB$

Q) $f(A,B,C) = \Sigma m(1,3,5,6,7)$

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	$_0$	1_1	1_3	$_2$
A 1	$_4$	1_5	1_7	1_6

$f(A,B,C) = C + AB$

Q) $f(A,B,C) = \Sigma m(0,2,3,4,6)$

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	1_0	$_1$	1_3	1_2
A 1	1_4	$_5$	$_7$	1_6

$f(A,B,C) = \bar{C} + \bar{A}B$

Q) $f(A,B,C) = \Sigma m(0,1,6,7) + d(3,4,5)$

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	1_0	1_1	X_3	$_2$
A 1	X_4	X_5	1_7	1_6

~~$f(A,B,C) = \Sigma m(0,1,6,7)$~~
$f(A,B,C) = \bar{B} + A$

Q) $f(A,B,C) = \Pi M(0,1,2,3,4,7)$

A \ BC	$B+C$ ~~$\bar{B}\bar{C}$~~ 00	$B+\bar{C}$ ~~$\bar{B}C$~~ 01	$\bar{B}+\bar{C}$ ~~BC~~ 11	$\bar{B}+C$ ~~$B\bar{C}$~~ 10
A ~~$\bar{A}$~~ 0	0_0	0_1	0_3	0_2
$\bar{A}$ ~~A~~ 1	0_4	$_5$	0_7	$_6$

$f = A.(B+C).(\bar{B}+\bar{C})$

→ 4-variable K-map

n-variable $= 2^n$ minterms
$= 2^n$ cells in K-map

4-variable = 16 minterms
= 16 cells

$f(A,B,C,D)$
↑ MSB ↑ LSB

AB (MSB) \ CD (LSB)	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10	
$\bar{A}\bar{B}$ 00	0	1	3	2	$A+B$
$\bar{A}B$ 01	4	5	7	6	$A+\bar{B}$
AB 11	12	13	15	14	$\bar{A}+\bar{B}$
$A\bar{B}$ 10	8	9	11	10	$\bar{A}+B$
	$C+D$	$C+\bar{D}$	$\bar{C}+\bar{D}$	$\bar{C}+D$	

Q) $f(A,B,C,D) = \Pi M(3,7,10,11,15)$

AB (MSB) \ CD (LSB)	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10	
$\bar{A}\bar{B}$ 00	$_0$	$_1$	0_3	$_2$	$A+B$
$\bar{A}B$ 01	$_4$	$_5$	0_7	$_6$	$A+\bar{B}$
AB 11	$_{12}$	$_{13}$	0_{15}	$_{14}$	$\bar{A}+\bar{B}$
$A\bar{B}$ 10	$_8$	$_9$	0_{11}	0_{10}	$\bar{A}+B$
	$C+D$	$C+\bar{D}$	$\bar{C}+\bar{D}$	$\bar{C}+D$	

$f = (\bar{C}+\bar{D})(\bar{A}+B+\bar{C})$
5 literals

$= \Sigma m(0,1,2,4,5,6,8,9,12,13,14)$

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	1_0	1_1	$_3$	1_2
$\bar{A}B$ 01	1_4	1_5	$_7$	1_6
AB 11	1_{12}	1_{13}	$_{15}$	1_{14}
$A\bar{B}$ 10	1_8	1_9	$_{11}$	$_{10}$

$f = \bar{C} + \bar{A}\bar{D} + B\bar{D}$
5 literals

Q)

AB \ CD	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}\bar{B}$ 00	X 0	1 1	3	1 2
$\bar{A}B$ 01	4	1 5	X 7	6
AB 11	1 12	X 13	X 15	14
$A\bar{B}$ 10	X 8	9	11	X 10

→ Unnecessary

2 variable eliminate

1 Quad + 2 Pairs → 1 variable eliminate

⇒ 8 literals

$(2 + 2\times 3 = 8)$

Q) Map 2-variable K-map into 3-variable & 3-variable K-map into 4-variable

$A \rightarrow MSB$

$BC \rightarrow LSB$

B \ C	0	1
$\bar{A}$ 0	0	1 1
A 1	A 2	1 3

⇒

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	0	1 1	1 3	2
A 1	4	1 5	1 7	1 6

C + AB

x \ yz	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	0	1 1	1 ~~3~~	~~2~~
A 1	4	1 5	1 7	1 6

~~w~~x \ yz	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	0 0	0 1	1 3	$\bar{w}$ 2
A 1	1 4	w 5	0 7	0 6

wx \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{w}\bar{x}$ 00	0	1	1 3	1 2
$\bar{w}x$ 01	1 4	0 5	7	6
wx 11	1 12	1 13	15	14
$w\bar{x}$ 10	~~1~~ 8	~~1~~ 9	1 11	0 10

$y = x\bar{y}\bar{z} + \bar{w}\bar{x}y + \bar{x}yz$

→ Concept of even & odd blocks in K-map

	0	1
0	E	O
1	O	E

	00	01	11	10
00	E 0	O 1	E 3	O 2
01	O 4	E 5	O 7	E 6
11	E 12	O 13	E 15	O 14
10	O 8	E 9	O 11	E 10

	00	01	11	10
0	ØE	O	E	O
1	O	ØE	O	E

XOR = 1 if odd no. of 1's are present

XNOR = 1, if even " " 0's " "

A \ BC	00	01	11	10
1	0	1 1	3	1 2
0	1 4	5	1 7	6

$Y = A \oplus B \oplus C$

$= \Sigma m(1,2,4,7)$

Q) Calculate the ~~maxm~~ minimal SOP & POS expression

~~f(x,y,~~ $f(w,x,y,z) = \bar{w}(\bar{x}\bar{y} + \bar{x}y + xy\bar{z}) + \bar{x}\bar{z}(y+w)$

$f^d(w,x,y,z) = \bar{w}x(\bar{y}z + y\bar{z}) + wyz$

wx \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{w}\bar{x}$ 00	1 0	1 1	1 3	1 2
$\bar{w}x$ 01	4	X 5	1 7	X 6
wx 11	12	13	X 15	14
$w\bar{x}$ 10	1 8	9	X 11	1 10

$f = \bar{w}\bar{x}\bar{y} + \bar{w}\bar{x}y + \bar{w}xy\bar{z} + \bar{x}y\bar{z} + w\bar{x}\bar{z}$

$f^d = \bar{w}x\bar{y}z + \bar{w}xy\bar{z} + wyz$

$f = \Sigma m(2,3,0,1,7,10,8)$

$f^d = \Sigma m(5,6,11,15)$

$f = \bar{w}\bar{x} + yz + \bar{x}\bar{z}$

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→ Logic operation using K-map

AND $f = f_1 \cdot f_2$

$f_1 = \Sigma m(0,1,2,4,6)$
$f_2 = \Sigma m(3,5,6,7,11,14)$
$f = \Sigma m(6)$

For AND operation we consider common minterms

OR $f = f_1 + f_2$

$f = \Sigma m(0,1,2,3,4,5,6,7,11,14)$
$f =$ union of ~~un~~ minterms

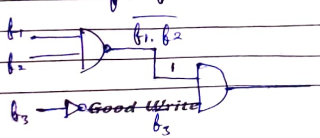
XOR $f = f_1 \oplus f_2 = \Sigma m(0,1,2,3,4,5,7,11,14)$

Consider uncommon minterms

Don't care

$1 \cdot X = X$	$0 \cdot X = 0$
$1 + X = 1$	$0 + X = X$
$1 \oplus X = \bar{X}$	$0 \oplus X = X$
$X + X = X$	$X \odot 0 = \bar{X}$
$X \cdot X = X$	$X \odot 1 = X$
$X \oplus X = X$	$X \odot X = X$

Q) Consider the circuit shown below & find the value of f_3



$f_1 = \Sigma m(0,1,3,5)$
$f_2 = \Sigma m(6,7)$
$f_3 = ?$ $f_x = \Sigma m(1,4,5)$

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$f_1 \cdot f_2 =$ Intersection of minterms
$= 0$
$\overline{f_1 \cdot f_2} = 1$

$f = \overline{f_3} = \Sigma m(1,4,5)$
$f_3 = \overline{\Sigma m(1,4,5)}$
$= \Sigma m(0,2,3,6,7)$

Q) $f(a,b,c) = ab + bc + ca$
$f[f(a,\bar{b},\bar{c})', f(\bar{a},b,c), c] = ?$

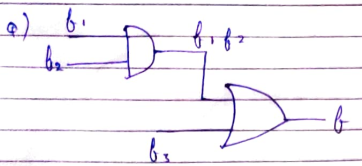
$x = f(a,\bar{b},\bar{c}) = \overset{10X}{a\bar{b}} + \overset{X00}{\bar{b}\bar{c}} + \overset{1X0}{a\bar{c}}$
$= \Sigma m(4,5,0,6)$
$= \Sigma m(0,4,5,6)$

$y = f(\bar{a},b,c) = \underset{01X}{\bar{a}b} + \underset{X11}{bc} + \underset{0X1}{\bar{a}c}$
$= \Sigma m(2,3,7,1)$
$= \Sigma m(1,2,3,7)$

$z = f = \overset{XX1}{c} = \Sigma m(1,3,5,7)$

$f = xy + yz + zx$
$xy = \Sigma m\ 0$
$yz = \Sigma m(1,3,7)$
$zx = \Sigma m(5)$

$f = \Sigma m(1,3,5,7) = c$

Q) 

$f_1 = \Sigma m(4,5,6,7,8)$
$f_2 = \Sigma m(1,6,15)$
$f_3 = ?$ $f = \Sigma m(1,6,8,15)$

$X = f_1 f_2 = \Sigma m(6)$
$f = X + f_3$
X & f_3 union $= \Sigma m(1,6,8,15)$
$\Sigma m(6) \cup f_3 = \Sigma m(1,6,8,15)$
$f_3 = \Sigma m(1,8,15)$
$= \Sigma m(1,6,8,15)$

→ 5 variable K-map

5 variable $= 2^5$ minterms $= 32$ minterms $= 32$ cells

$f(A,B,C,D,E) = A.\underbrace{f(1,B,C,D,E)}_{16 \text{ cells}} + \bar{A}.\underbrace{f(0,B,C,D,E)}_{16 \text{ cells}}$

K-map-1: 10000 → 11111, 16 to 31

K-map 2: 00000 → 01111, 0 to 15

Q) $f(A,B,C,D,E) = \Sigma m(0,2,3,10,11,12,15,16,17,18,19,20,21,26,27)$

A = 0

BC \ DE	$\bar{D}\bar{E}$ 00	$\bar{D}E$ 01	DE 11	$D\bar{E}$ 10
$\bar{B}\bar{C}$ 00	1 (0)	1	1 (3)	1 (2)
$\bar{B}C$ 01	4	5	7	6
BC 11	1 (12)	13	1 (15)	14
$B\bar{C}$ 10	8	9	1 (11)	1 (10)

A = 1

BC \ DE	00	01	11	10
$\bar{B}\bar{C}$ 00	1 (16)	1 (17)	1 (19)	1 (18)
01	1 (20)	1 (21)	23	22
11	28	29	31	30
10	24	25	1 (27)	1 (26)

Adjacent

Adjacent

~~$f = \bar{B}\bar{C}D + \bar{B}C\bar{D}$~~ $f = \bar{C}D + \bar{B}\bar{D}A + \bar{B}\bar{C}\bar{E}\bar{A} + BDE\bar{A} + BC\bar{D}\bar{E}\bar{A}$

18 literals

Q) $f(A,B,C,D,E) = AB + CD + \bar{A}\bar{B}E + B\bar{C}\bar{D}\bar{E}$

A = 0

BC \ DE	$\bar{D}\bar{E}$ 00	$\bar{D}E$ 01	DE 11	$D\bar{E}$ 10
$\bar{B}\bar{C}$ 00		1	1	
$\bar{B}C$ 01		1	1	1
BC 11			1	1
$B\bar{C}$ 10	1			

A = 1

BC \ DE	00	01	11	10
00				
01			1	1
11	1	1	1	1
10	1	1	1	1

$f = BA + CD + \bar{B}EA + B\bar{C}\bar{D}\bar{E}$

→ Prime Implicant

Implicant

A \ BC	00	01	11	10
0	1	1	1	1
1		1	1	

$m_0, m_1, m_2, m_3, m_5, m_7$
Implicant

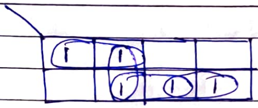
Pairs: 7
Quad: 2 } Implicant

Bade ke andar chhota count mat karna

Prime

A \ BC	00	01	11	10
0	1	1	1	1
1		1	1	

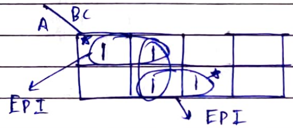
$\boxed{P.I = 2}$



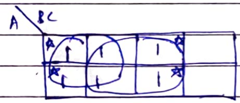
PI = 4

→ Essential PI (EPI)

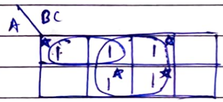
It is PI in which there is atleast 1 minterm which is not covered by any PI.



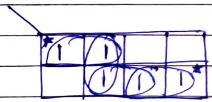
EPI = 2



EPI = 2



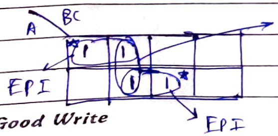
EPI = 2



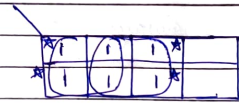
EPI = 2

→ Redundant PI (RPI)

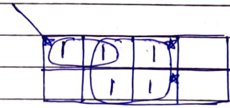
The PI in which each minterm is covered by an EPI is called as RPI



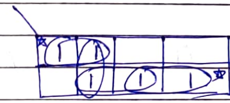
RPI = 1



RPI = 0



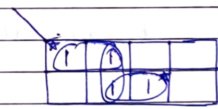
RPI = 0



RPI = 0

→ Selective PI (SPI)

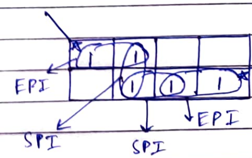
The prime implicants in which all minterms are covered by multiple PI but there is atleast 1 minterm which is not the part of other EPI



PI = 3 RPI = 1

EPI = 2 SPI = 0

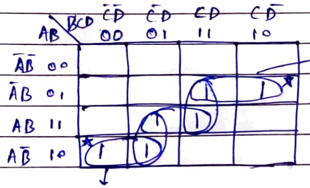
$\boxed{PI = EPI + RPI + SPI}$



EPI = 2

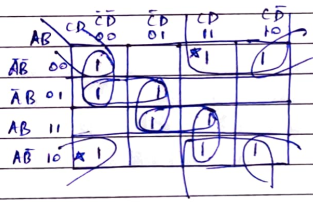
SPI = 2

A minimum logic expression always contains some SPI & all EPI



EPI = 2
RPI = 0
SPI = 3

$A\bar{B}\bar{C} + \bar{A}BC + ABD$



PI = 5 pair + 2 quad = 7
EPI = 2 quad = 2
RPI = 0
SPI = 5

$\bar{B}C + \bar{B}\bar{D} + ABD + \bar{A}B\bar{C}$

→ 2421 Code → BCD code

Set 2 ← Self complementing prop. of 2421 code

Decimal Digit	Set 1 (2421)	Set 2 (2421)
0	0000	0000
1	0001	0001
2	0010	0010
3	0011	0011
4	0100	0100
5	0101	1011
6	0110	1100
7	0111	1101
8	1110	1110
9	1111	1111

Q) XS-3 code for decimal nos.

i) 24

0010 0100
0011 0011
0101 0111 XS-3 code for 24
5 7

→ Binary to Gray Code

Q) Convert 1011 to Gray code

MSB → 1 0 1 1 ← LSB
+1 +1 +1 +1
1 1 1 0 0

1110

DATE: ___/___/_____
PAGE: _________

Q) Convert 1110 to BCD

1 1 1 0

$\boxed{1\ 0\ 1\ 1}$

→ Hamming Code - Error detection

7 bit Hamming Code is used commonly

Data bits - 4
Parity bits - 3 $\quad 2^m$ {where $m = 0, 1, ----, m$}

$2^0 = 1$ (P_1)
$2^1 = 2$ (P_2)
$2^2 = 4$ (P_4)
$2^3 = 8$ (X)

D_7	D_6	D_5	P_4	D_3	P_2	P_1
7	6	5	4	3	2	1

$P_1 \rightarrow D_3\ D_5\ D_7$

$P_2 \rightarrow D_3\ D_6\ D_7$

$P_4 \rightarrow D_5\ D_6\ D_7$

Q) 1 0 1 1

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	0	1		1		

$P_1 = D_3\ D_5\ D_7 = 0\ 1\ 1\ 1$

$P_1 = 1$ (To make it even)

$P_2 = D_3\ D_6\ D_7 = 0\ 1\ 0\ 1$

$P_2 = 0$

$P_4 = 0\ 1\ 0\ 1$

$P_4 = 0$

1010101

DATE: ___/___/_____
PAGE: _________

Q) If 7-bit hamming code word received by a receiver is 1011011. Assuming even parity state ~~wether~~ whether the received code word is incorrect or correct. If wrong code locate the bit having error

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	0	1	1	0	1	1

$P_4 = D_5\ D_6\ D_7$
1 1 0 1 (3) → Odd
$\boxed{P_4 = 1}$ ★

$P_2 = D_3\ D_6\ D_7$
1 0 0 1 (2) → Even ✓
$\boxed{P_2 = 0}$

$P_1 = D_3\ D_5\ D_7$
1 0 1 ~~0~~1 (3) → Odd
$\boxed{P_1 = 0}$

$P_4\ P_2\ P_1 = (1\ 0\ 1)_2 = (5)_{10}$
↓
Bit error

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	0	1	1	0	1	1

(1 0 0 1 0 1 1) → Correct code

→ Quine - McCluskey minimization method (Tabular method)

Q) $Y(A, B, C, D) = \Sigma m(0, 1, 3, 7, 8, 9, 11, 15)$

Step-1 Group	Minterm	Bin. rep. A B C D
0	m_0	0 0 0 0
1	m_1	0 0 0 1
	m_8	1 0 0 0
2	m_3	0 0 1 1
	m_9	1 0 0 1
3	m_7	0 1 1 1
	m_{11}	1 0 1 1
4	m_{15}	1 1 1 1

Step-2

Group	Matched pairs	Binary Rep. A B C D
0	$m_0 - m_1$	0 0 0 _
	$m_0 - m_8$	_ 0 0 0
1	$m_1 - m_3$	0 0 _ 1
	$m_1 - m_9$	_ 0 0 1
	$m_8 - m_9$	1 0 0 _
2	$m_3 - m_7$	0 _ 1 1
	$m_3 - m_{11}$	_ 0 1 1
	$m_9 - m_{11}$	1 0 _ 1
3	$m_7 - m_{15}$	_ 1 1 1
	$m_{11} - m_{15}$	1 _ 1 1

Step-3

Group	Matched pairs	B.R A B C D	
0	$m_0 - m_1 - m_8 - m_9$	_ 0 0 _	$\bar{B}\bar{C}$
	$m_0 - m_8 - m_1 - m_9$	_ 0 0 _	
1	$m_1 - m_3 - m_9 - m_{11}$	_ 0 _ 1	$\cancel{\bar{A}}\bar{B}D$
	$m_1 - m_9 - m_3 - m_{11}$	_ 0 _ 1	
2	$m_3 - m_7 - m_{11} - m_{15}$	_ _ 1 1	CD
	$m_3 - m_{11} - m_7 - m_{15}$	_ _ 1 1	

PI

PI	Minterms	0	1	3	7	8	9	11	15
$\bar{B}\bar{C}$	0, 1, 8, 9	X	X			X	X		
$\bar{B}D$	1, 3, 9, 11		X	X			X	X	
CD	3, 7, 11, 15			X	X			X	X

$$\boxed{Y = \bar{B}\bar{C} + CD}$$